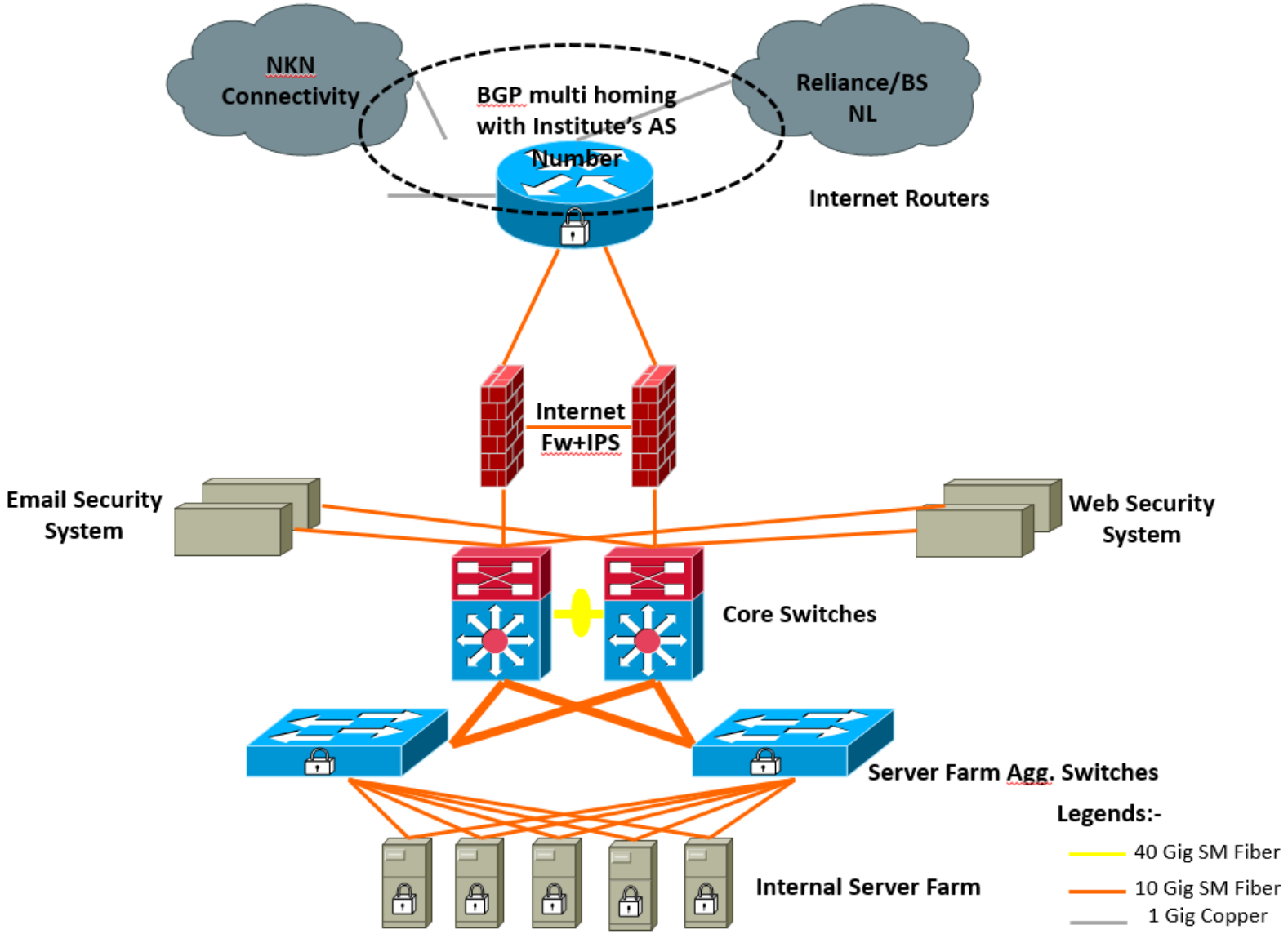
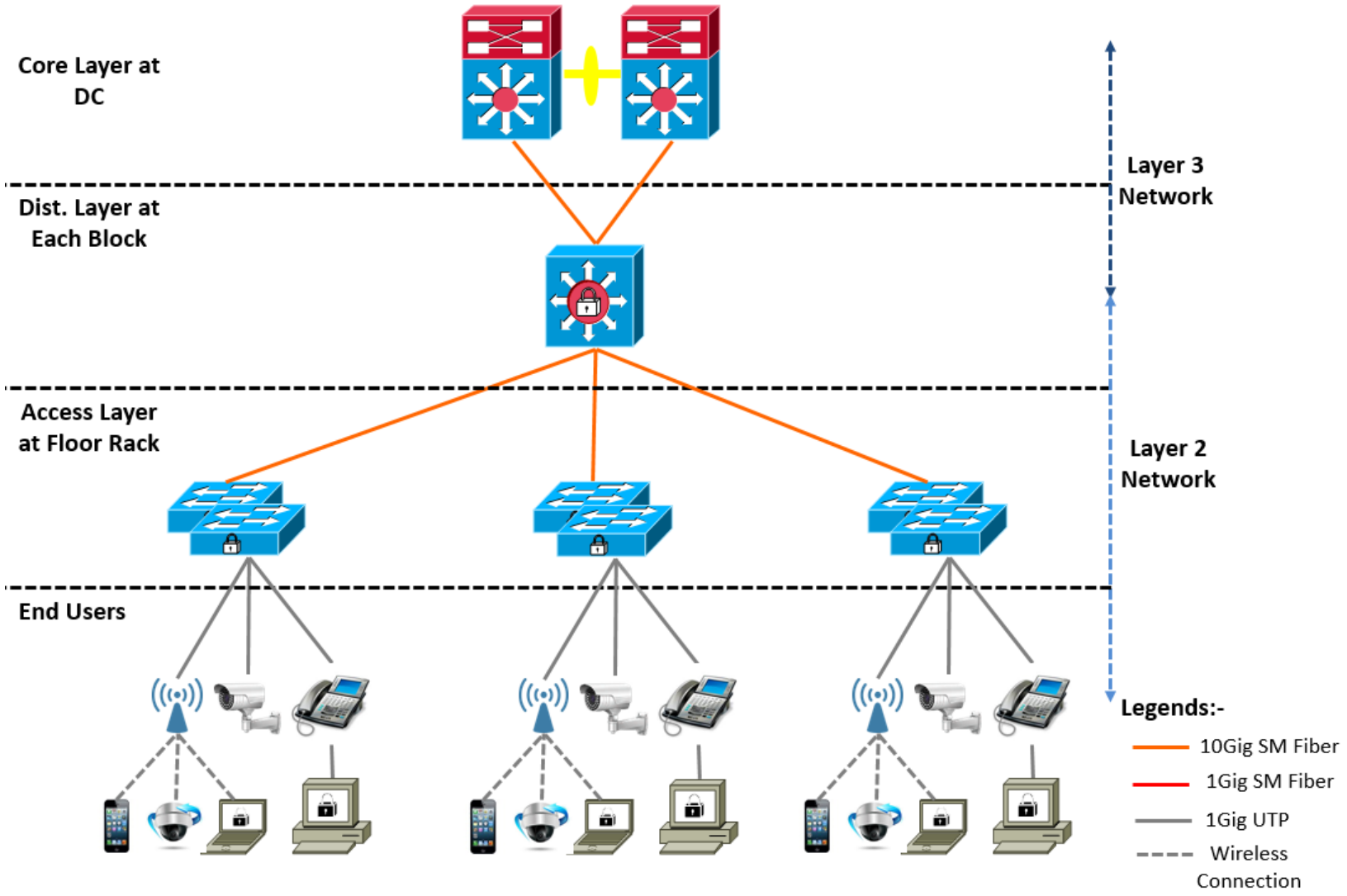
Network Architecture at DC



Access Network Architecture





Software Defined Networks and OpenFlow Protocol

Anand Baswade

anand@iitbhilai.ac.in

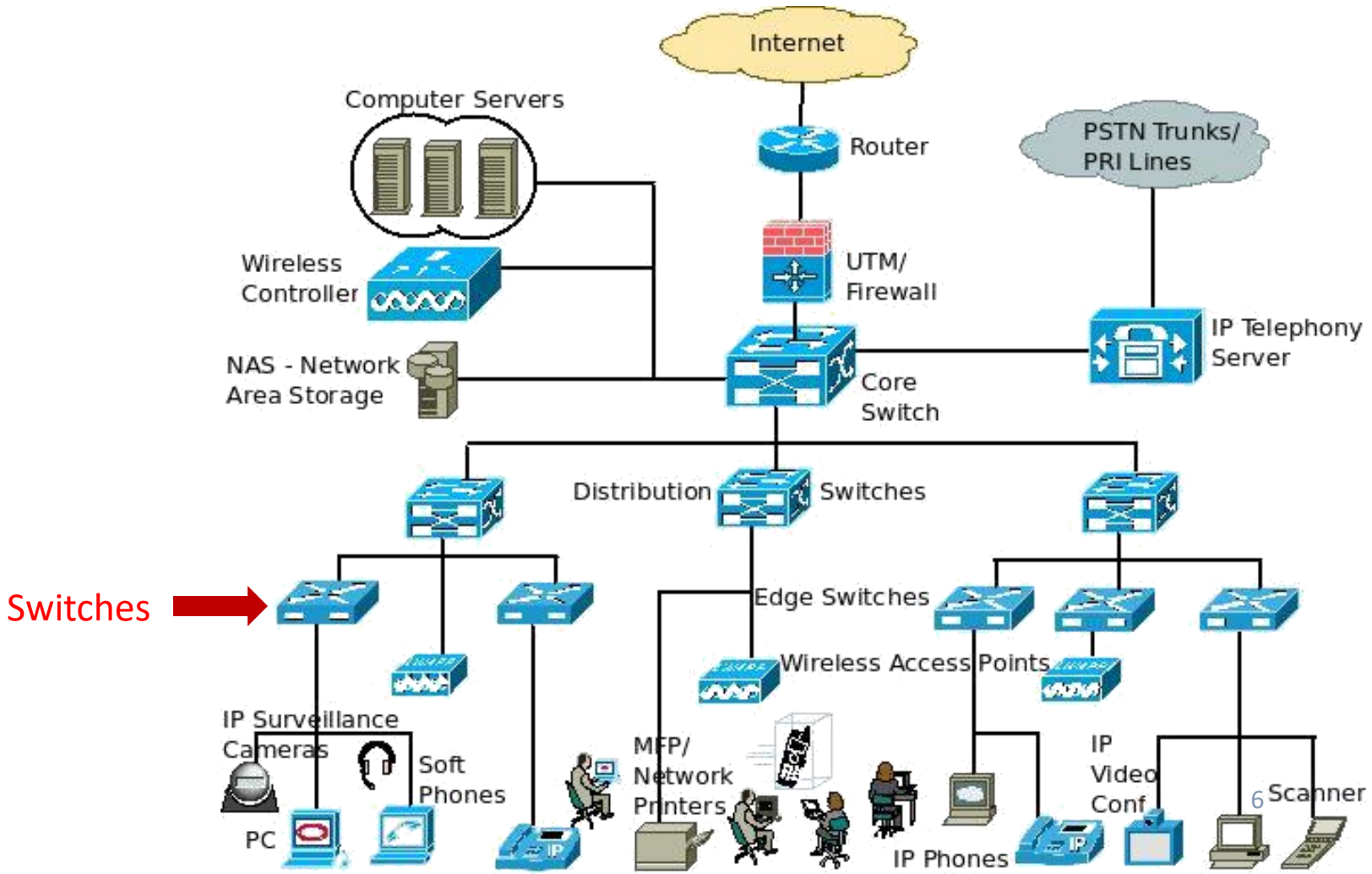
Source

- Software-Defined Networking: A Comprehensive Survey by Diego Kreutz and others
- SDN and OpenFlow Slides from Saurav Das With contributions Nick McKeown, Guru Parulkar, Scott Shenker, Brandon Heller, Rob Sherwood, Guido Appenzeller, Martin Casado and many, many others…
- Talk by Prof. Scott Shenkar from UC berkeley : https://www.youtube.com/watch?v=WVs7Pc99S7w
- SDN slides @ Duke

Outline

- **What is SDN?**
  - Background
  - An OS for networks
- **What is OpenFlow?**
  - How it helps SDN

Limitations of Current Networks

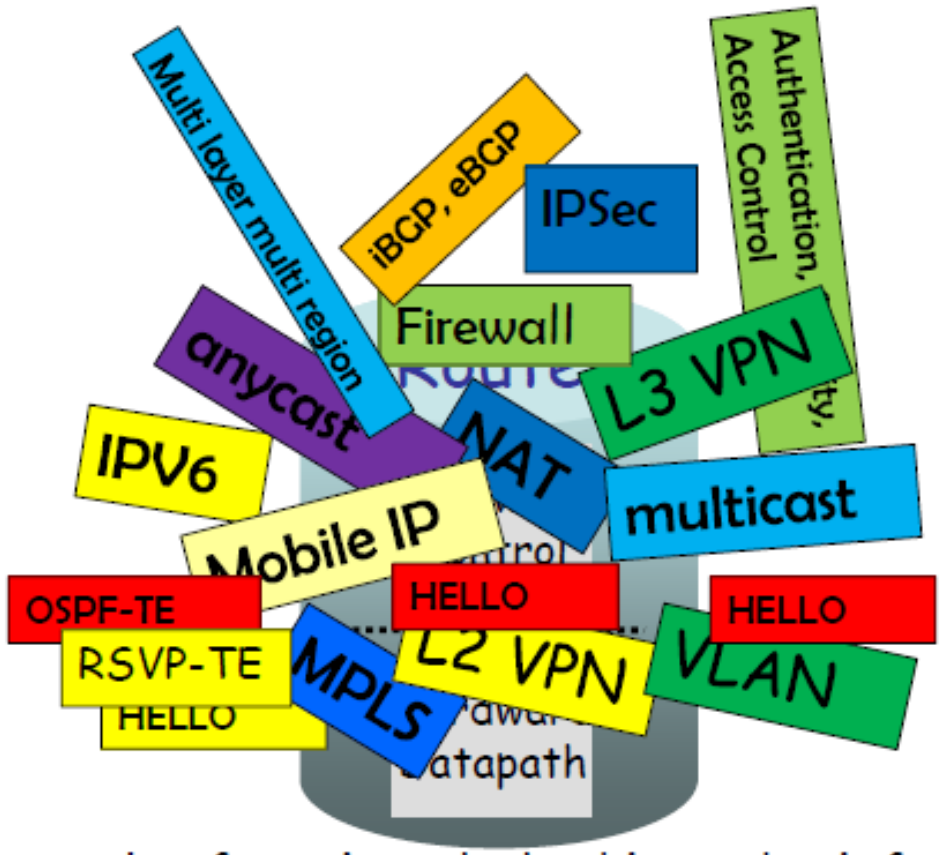


http://www.excitingip.net/27/a-basic-enterprise-lan-network-architecture-block-diagram-and-components/

Networks are getting Complex

- Networks used to be simple
  - Basic Ethernet/IP straightforward, easy to manage
- New control requirements have led to complexity
  - Access Control Lists, VLANs, Traffic Engineering, Middleboxes, DPI, ….
- The infrastructure still works … (Why?)
  - Only because of your great ability to master complexity
- Extreme complexity often signifies weak foundation

Cont..

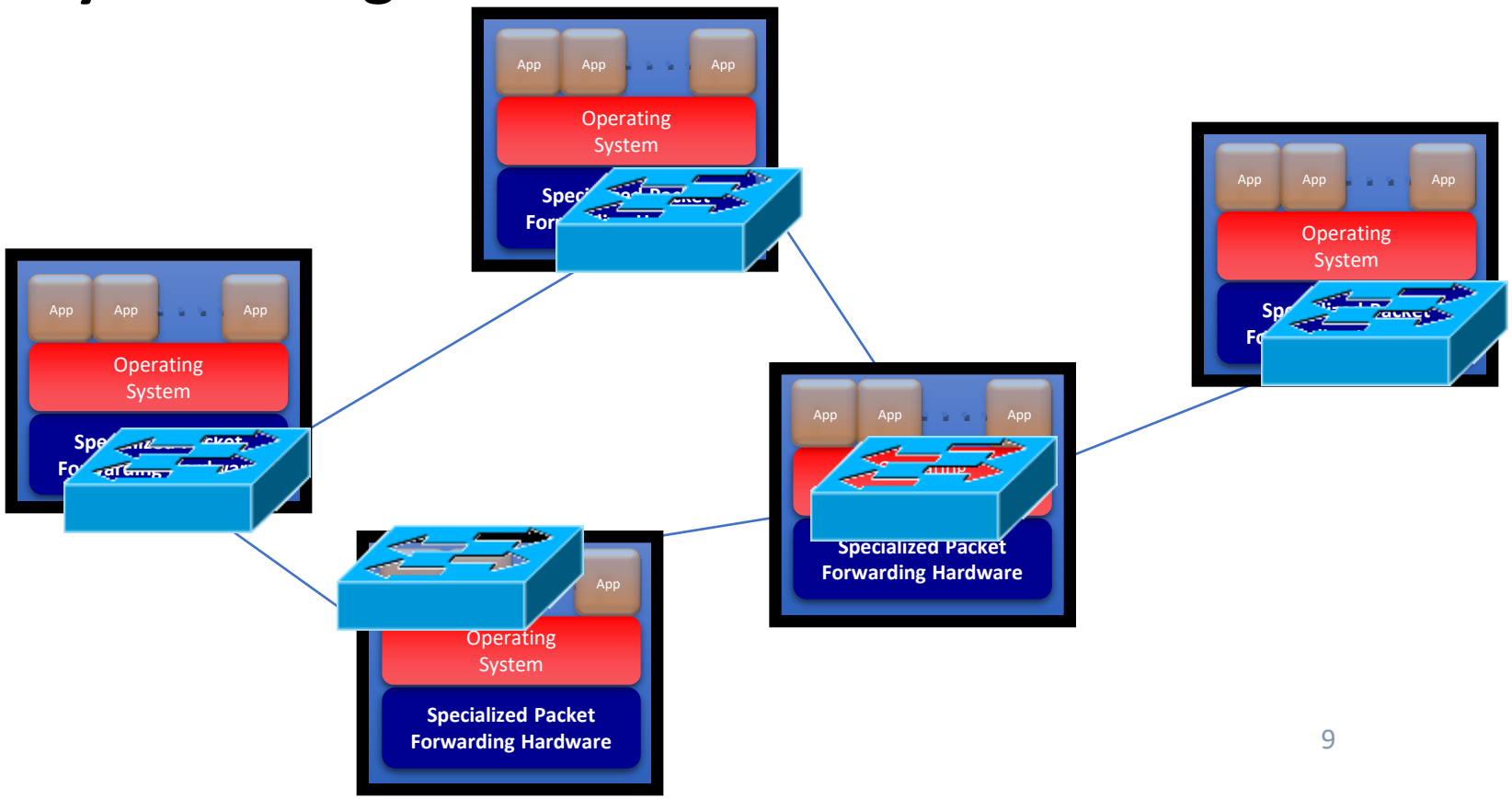


Many complex functions baked into the infrastructure

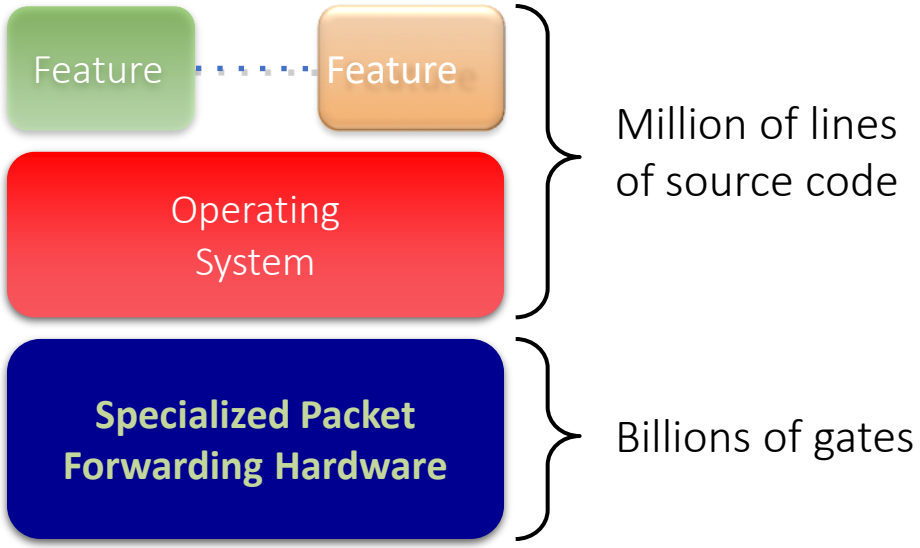
OSPF, BGP, multicast, differentiated services,
Traffic Engineering, NAT, firewalls, MPLS, redundant layers, …

Limitations of Current Networks

- **Old ways to configure a network**



Limitations of Current Networks



Many complex functions baked into infrastructure

OSPF, BGP, multicast, differentiated services, Traffic Engineering, NAT, firewalls, …

Cannot dynamically change according to network conditions

Limitations of Current Networks

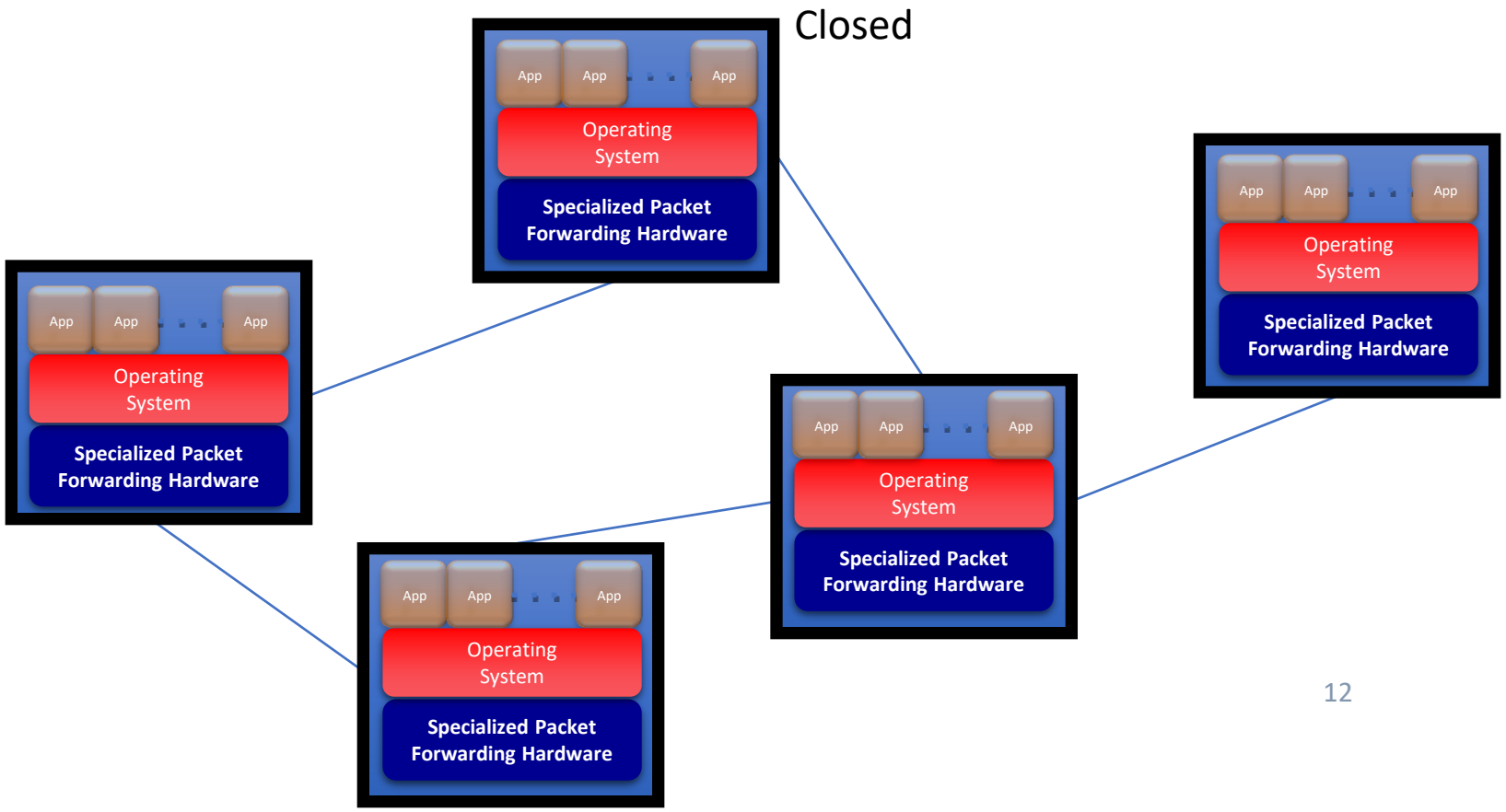
- **No control plane abstraction for the whole network!**

- **It's like old times – when there was no OS…**

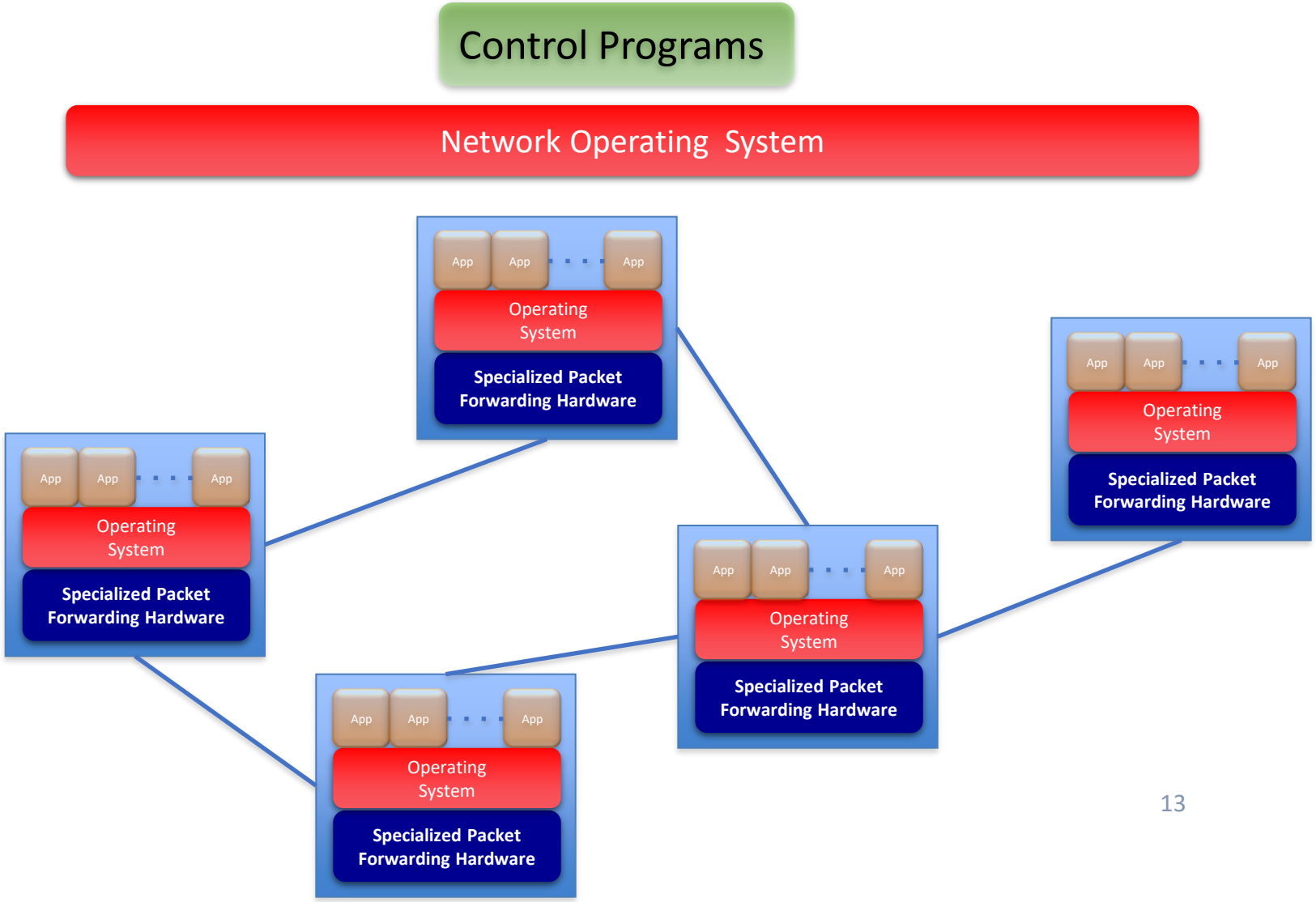


Wilkes with the EDSAC, 1949

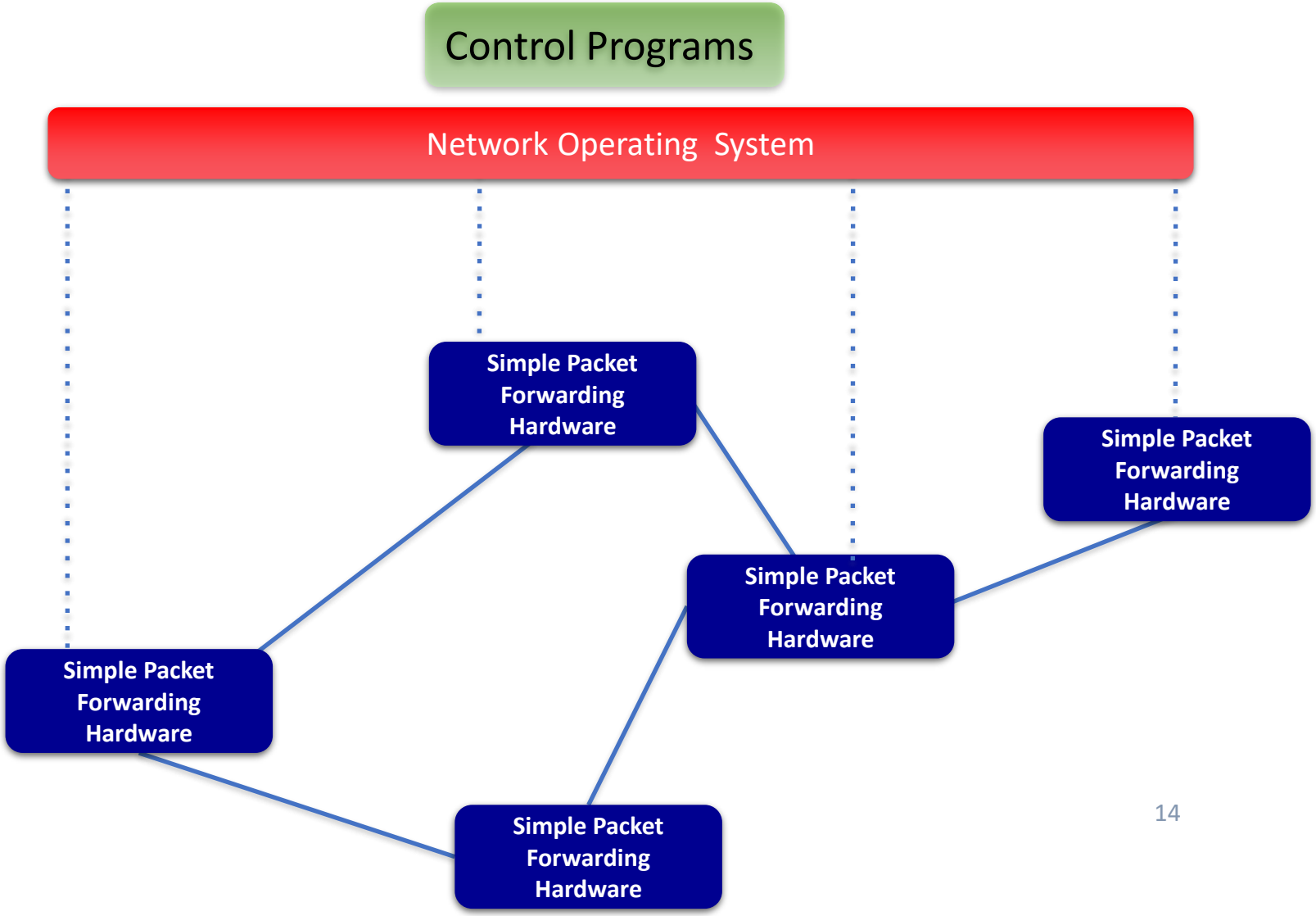
Idea: An OS for Networks



Idea: An OS for Networks



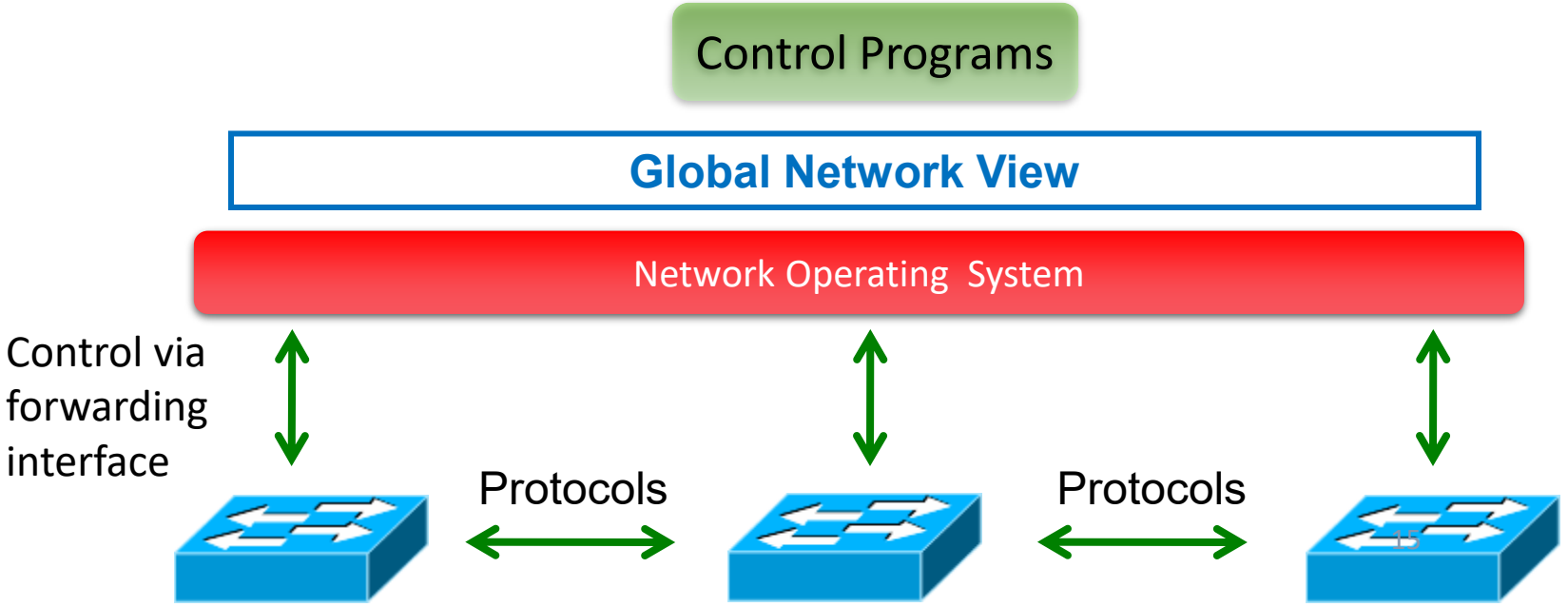
Idea: An OS for Networks



Idea: An OS for Networks

- **Towards an Operating System for Networks”**

Software-Defined Networking (SDN)



The Future of Networking, and the Past of Protocols, Scott Shenker, *with Martin Casado, Teemu Koponen, Nick McKeown*

Software Defined Networking

- **No longer designing distributed control protocols**

- **Much easier to write, verify, maintain, …**
  - An interface for programming

- **NOS serves as fundamental control block**
  - With a global view of network

SDN architecture and its fundamental abstractions

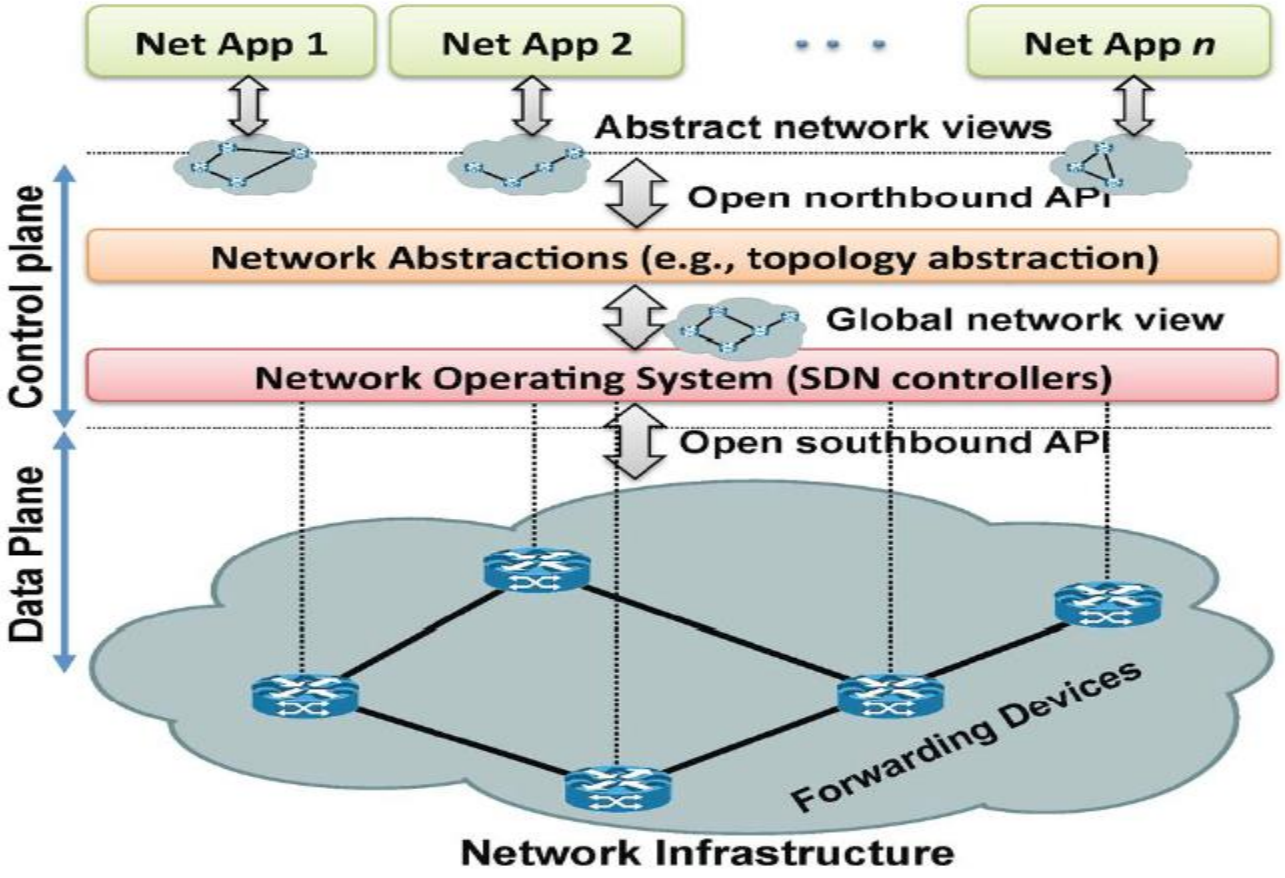
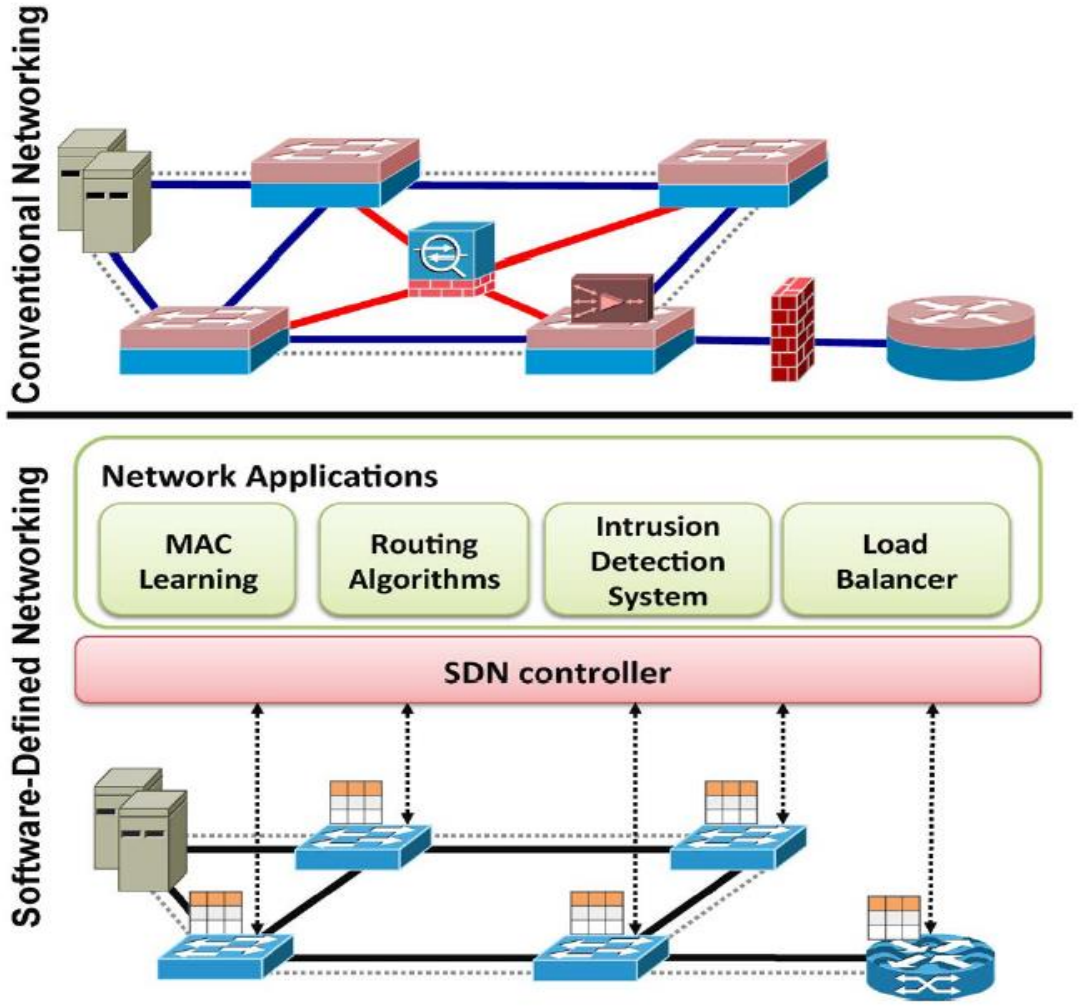


Fig. 4. *SDN architecture and its fundamental abstractions.*

Traditional networking versus SDN



Distributed controllers: east/westbound APIs

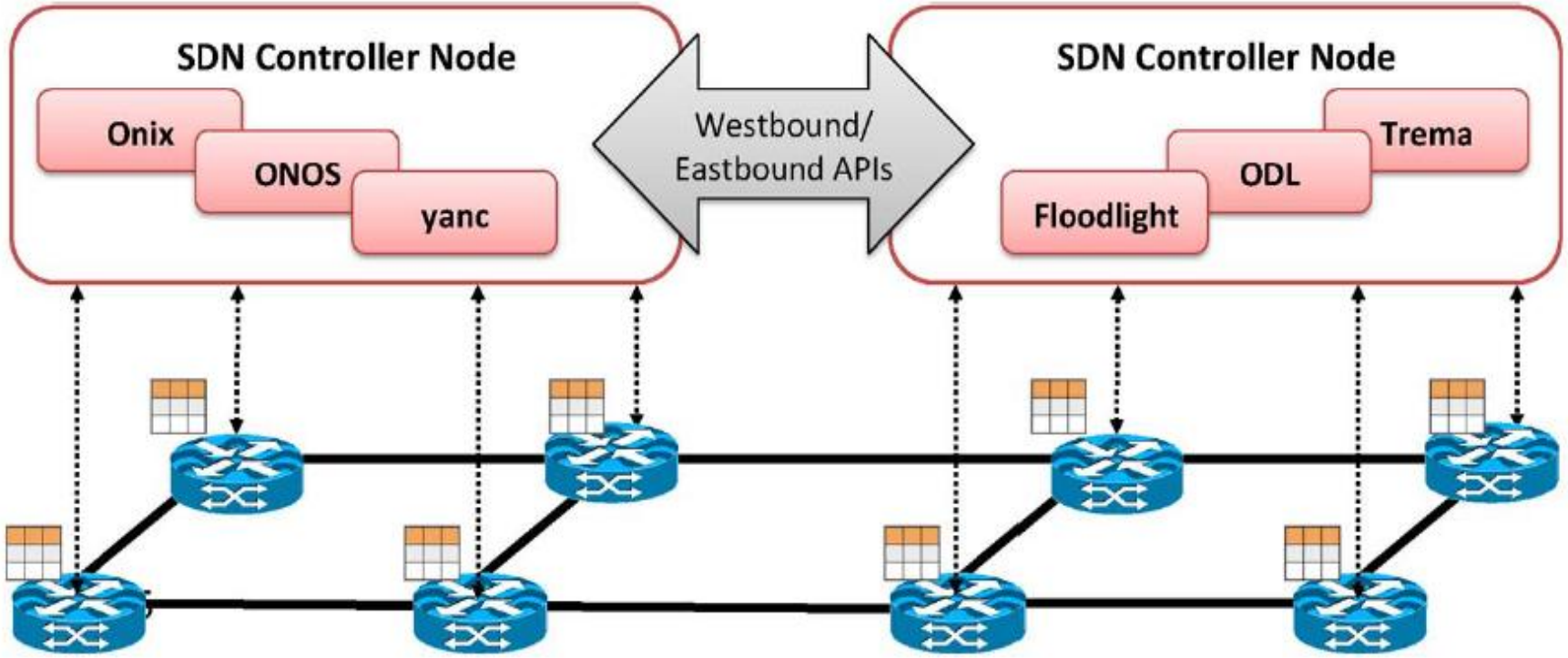


Fig. 9. ***Distributed controllers: east/westbound APIs.***

Standard Bodies

- Open Networking Foundation
  - http://www.openflow.org/
  - https://www.opennetworking.org/

- IETF
  - http://tools.ietf.org/html/draft-nadeau-sdn-problem-statement-00
  - http://tools.ietf.org/html/draft-nadeau-sdn-framework-0

Benefits of SDN

- Ease Innovation in Network
- Experiment and research using non-expensive equipment
- More accessibility since software can be easily developed
- Quick deployment – No hardware fabrication cycles
- More Flexibility with programmability
- Ease of customization and integration with other software applications
- Fast upgrades
- Program a network Vs Configure a network

Software Defined Networking

- **Questions:**
  - How to obtain global information?
  - What are the configurations?
  - How to implement?
  - How is the scalability?
  - How does it really work?

Outline

- **What is SDN?**
  - Limitations of current networks
  - The idea of Network OS

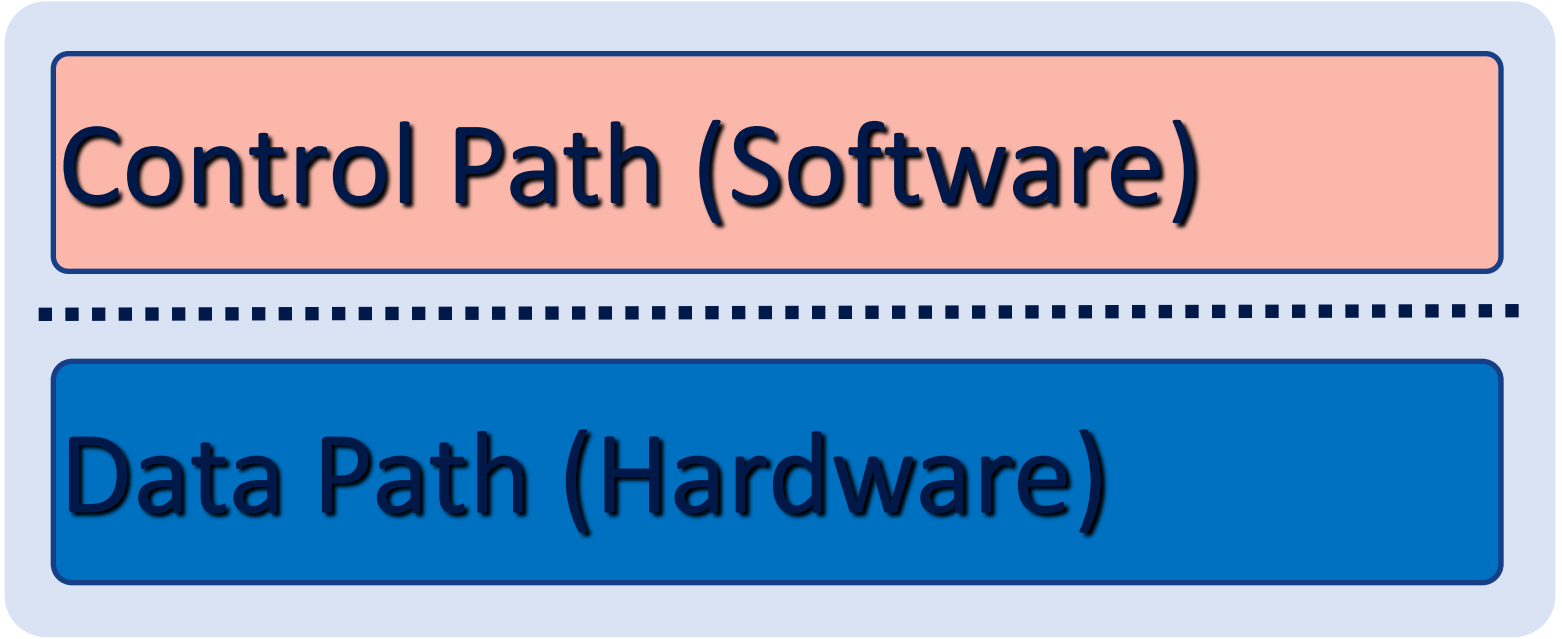


- **What is OpenFlow?**
  - How it helps SDN

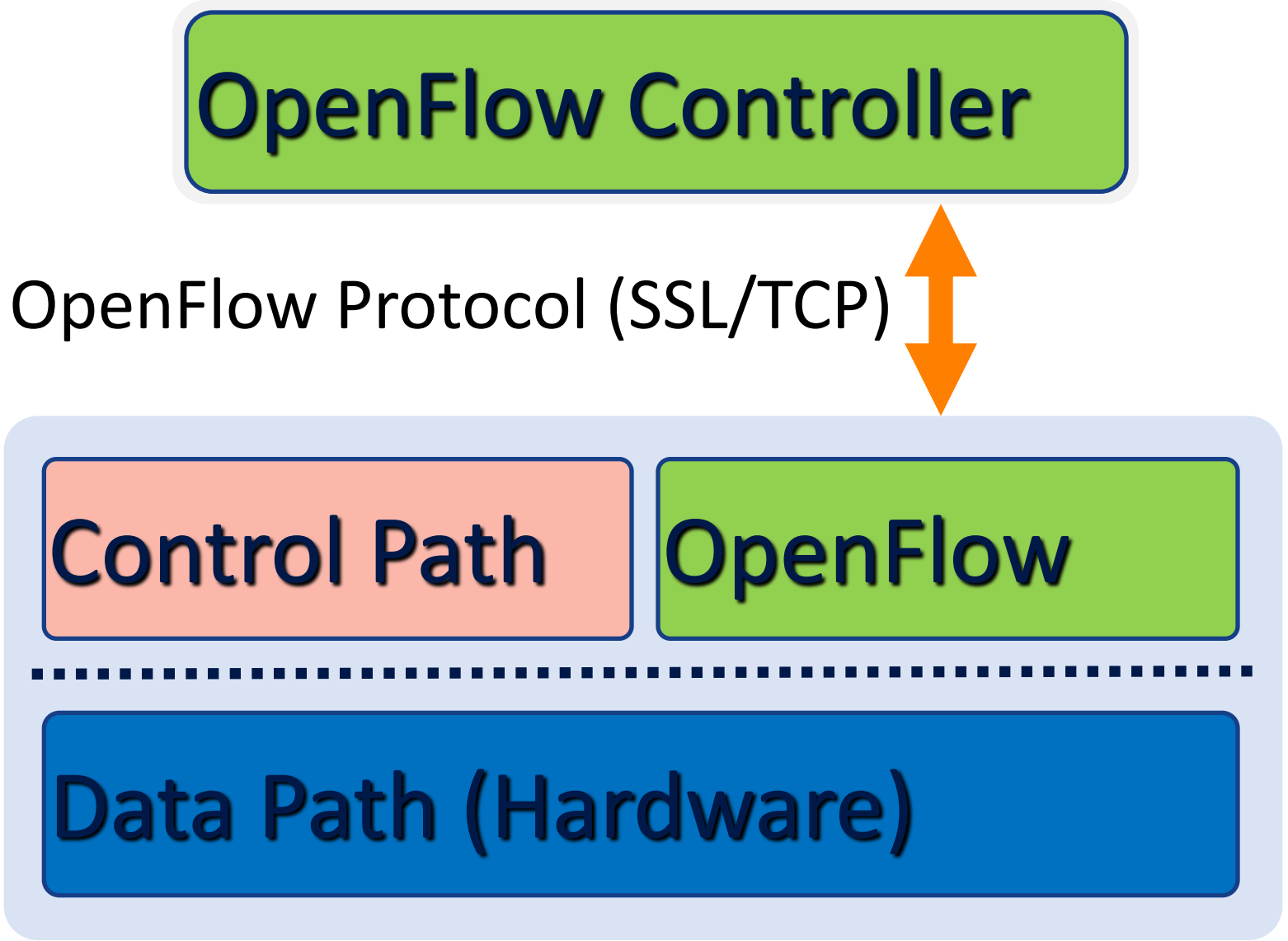
OpenFlow

- **Like hardware drivers**
  - interface between switches and Network OS

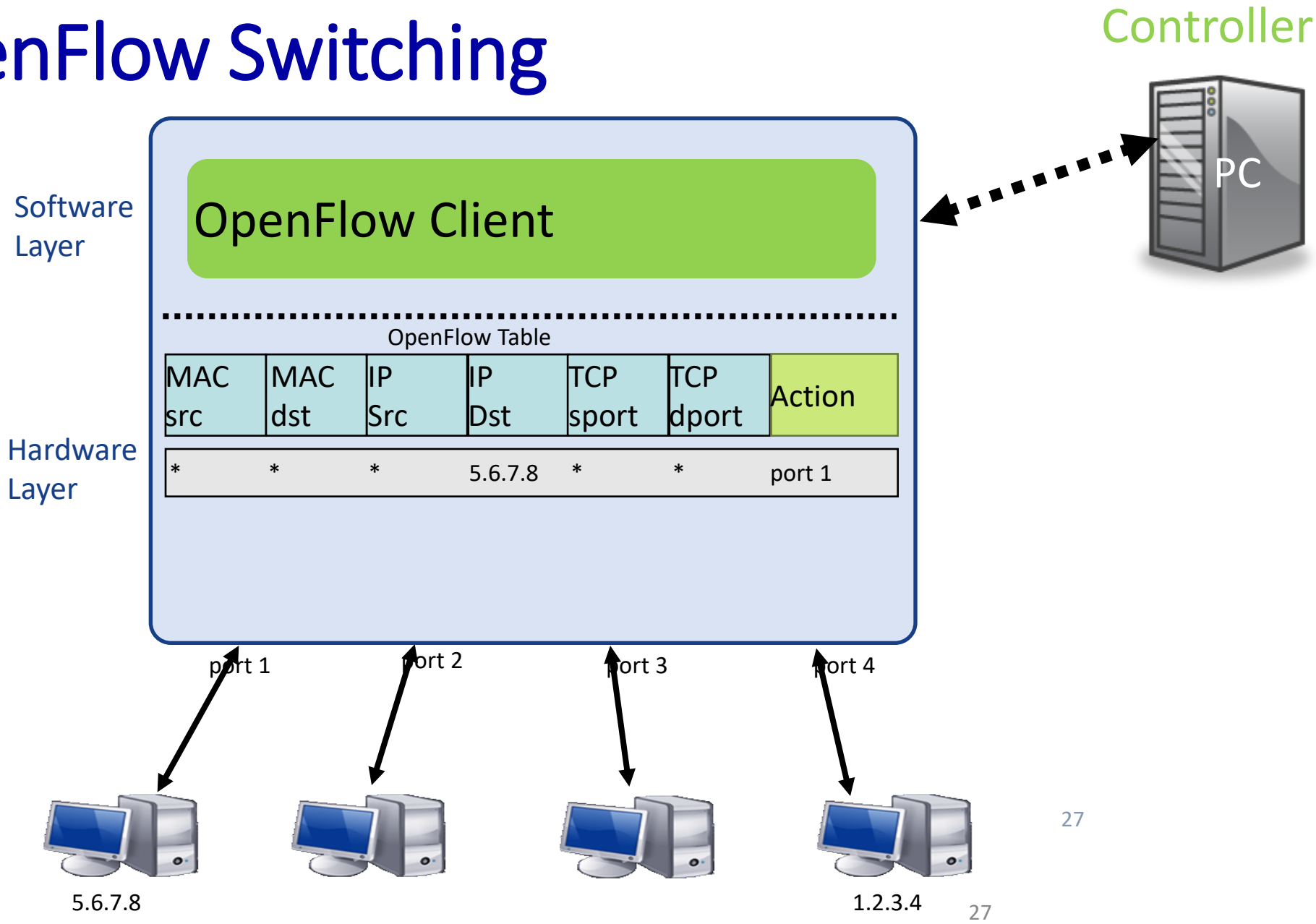
OpenFlow



OpenFlow

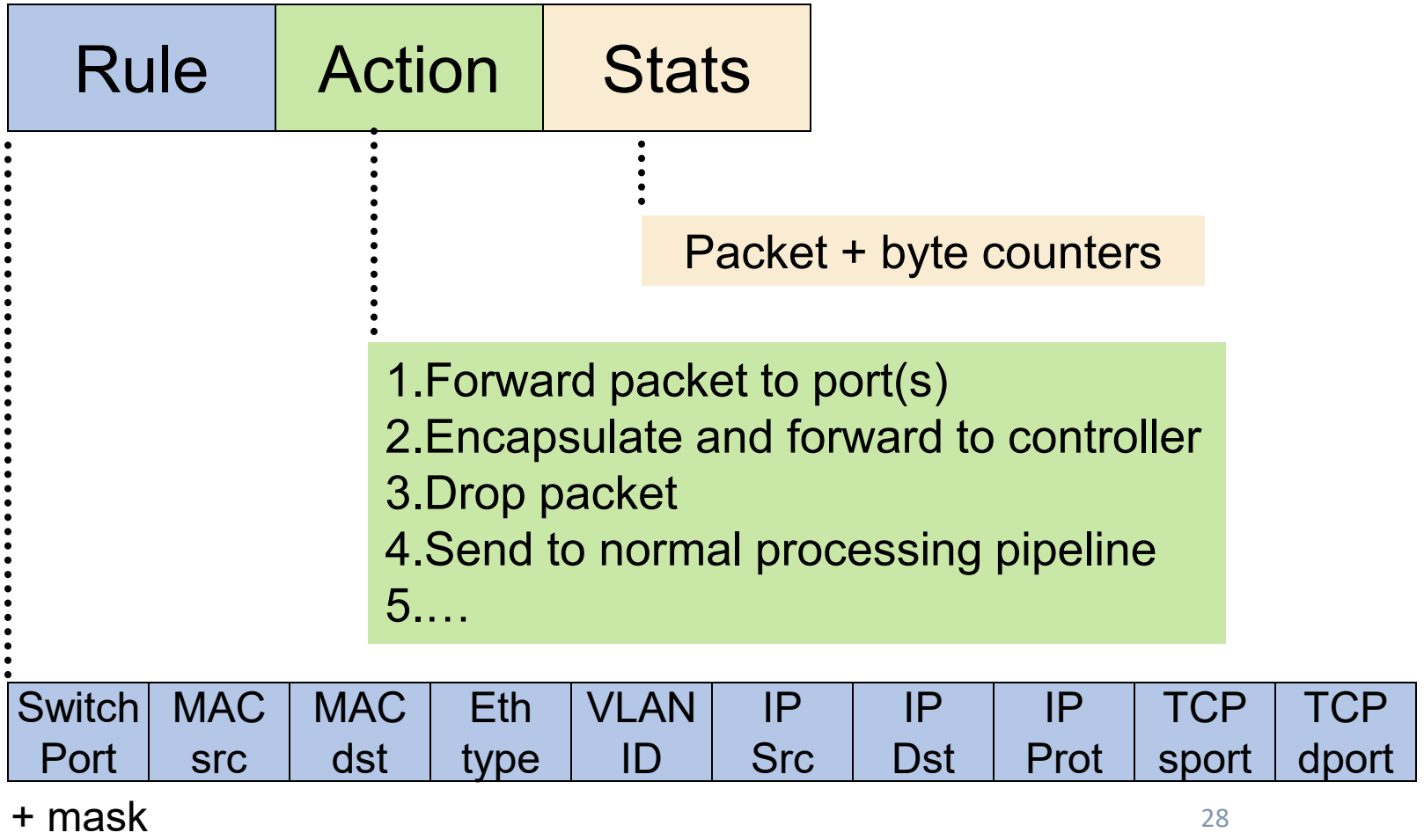

OpenFlow Controller
OpenFlow Protocol (SSL/TCP)
Control Path
OpenFlow
Data Path (Hardware)

OpenFlow Switching



27

OpenFlow Table Entry



OpenFlow Examples

Switching

Switch Port	MAC src	MAC dst	Eth type	VLAN ID	IP Src	IP Dst	IP Prot	TCP sport	TCP dport	Action
*	*	00:1f:..	*	*	*	*	*	*	*	port6

Routing

Switch Port	MAC src	MAC dst	Eth type	VLAN ID	IP Src	IP Dst	IP Prot	TCP sport	TCP dport	Action
*	*	*	*	*	*	5.6.7.8	*	*	*	port6

Firewall

Switch Port	MAC src	MAC dst	Eth type	VLAN ID	IP Src	IP Dst	IP Prot	TCP sport	TCP dport	Action
*	*	*	*	*	*	*	*	*	22	drop

OpenFlow-enabled SDN devices

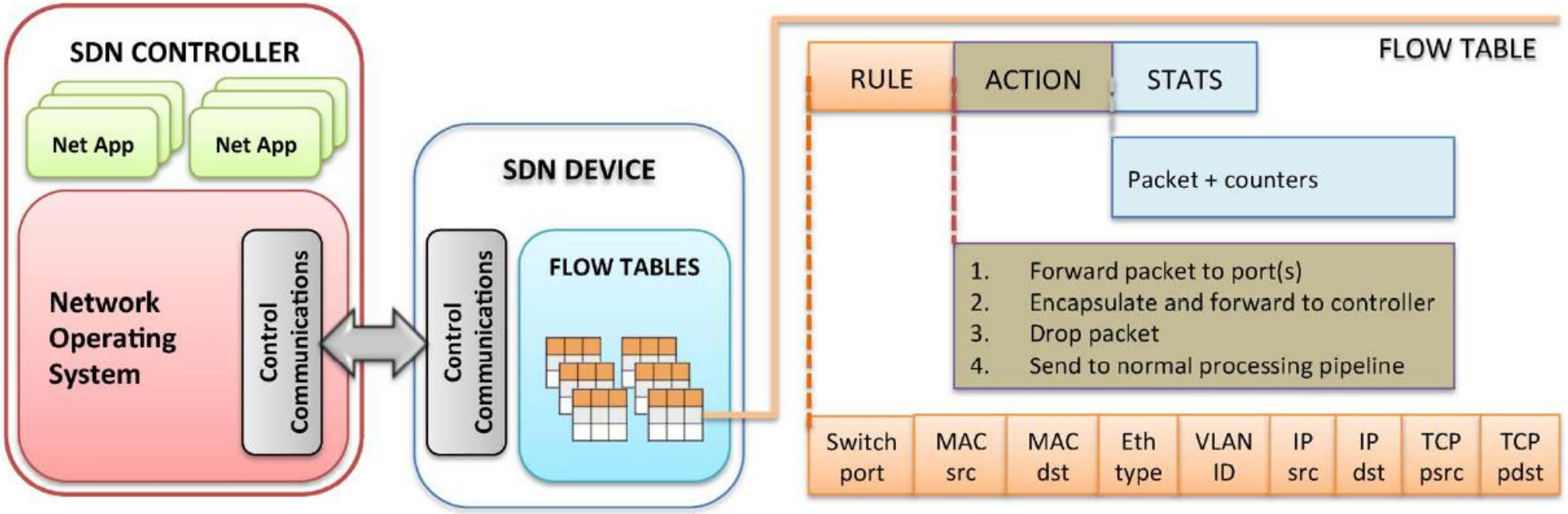
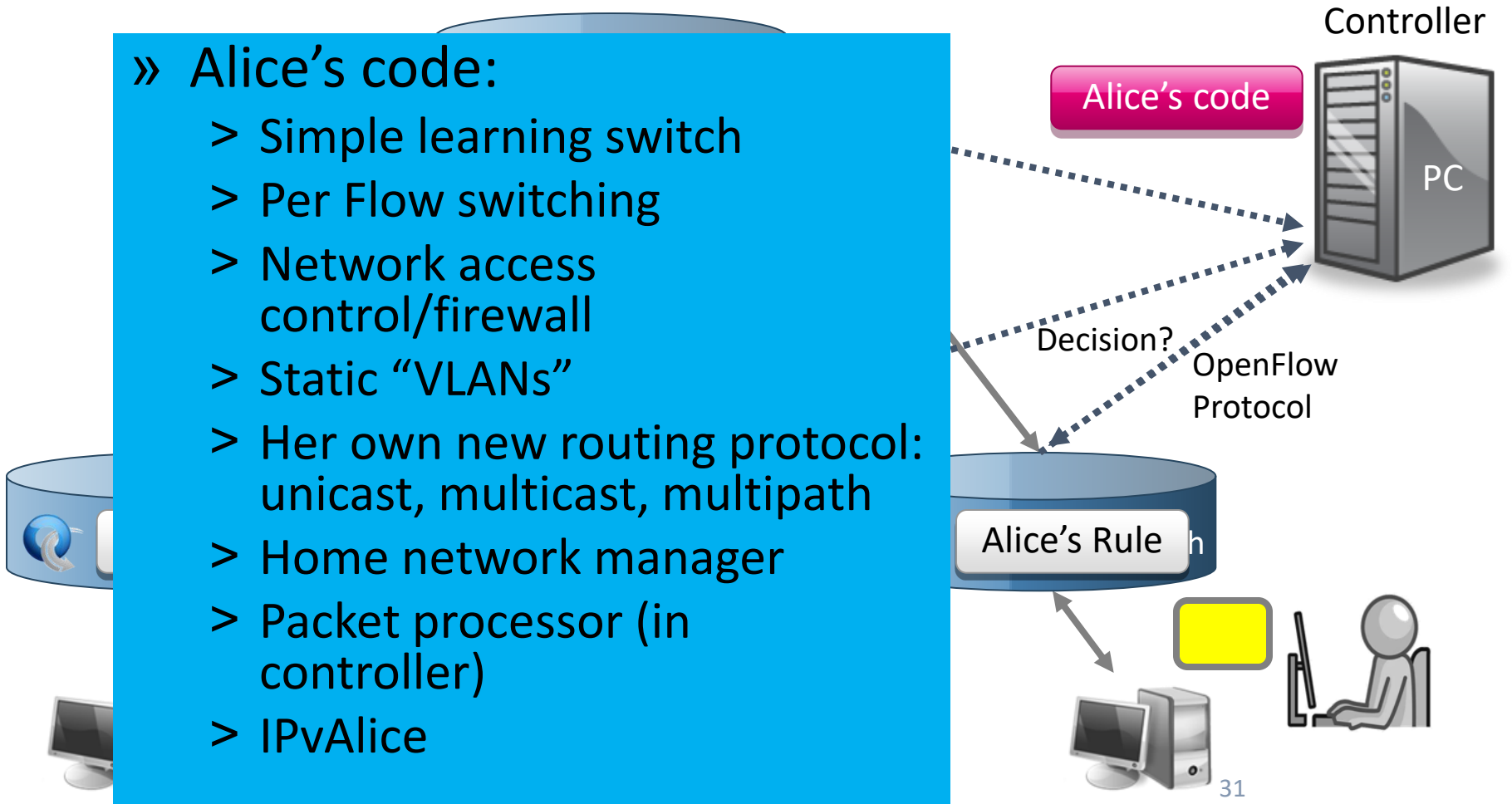


Fig. 7. ***OpenFlow-enabled SDN devices.***

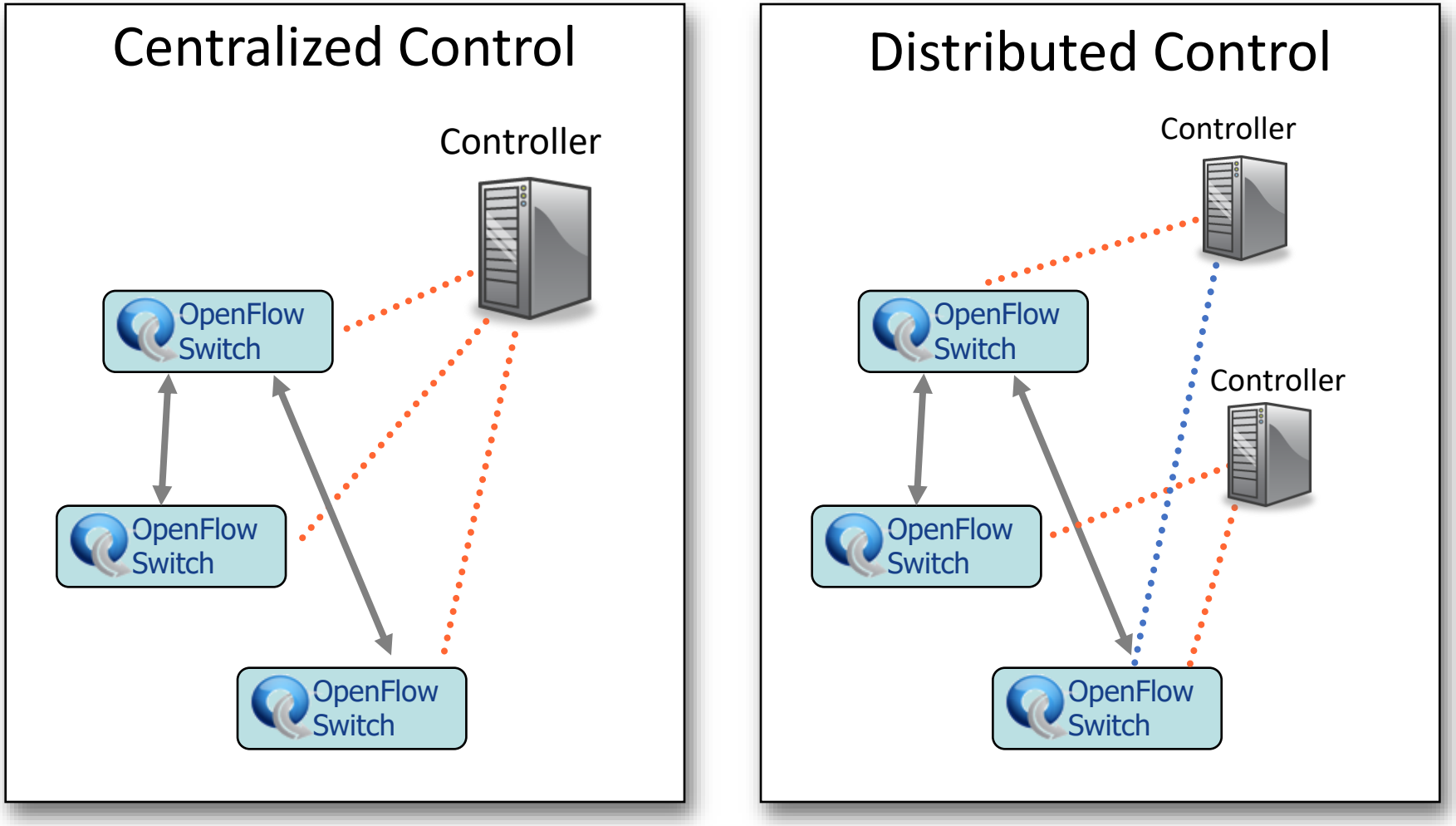
OpenFlow Usage

- Alice's code:
  - Simple learning switch
  - Per Flow switching
  - Network access control/firewall
  - Static "VLANs"
  - Her own new routing protocol: unicast, multicast, multipath
  - Home network manager
  - Packet processor (in controller)
  - IPvAlice



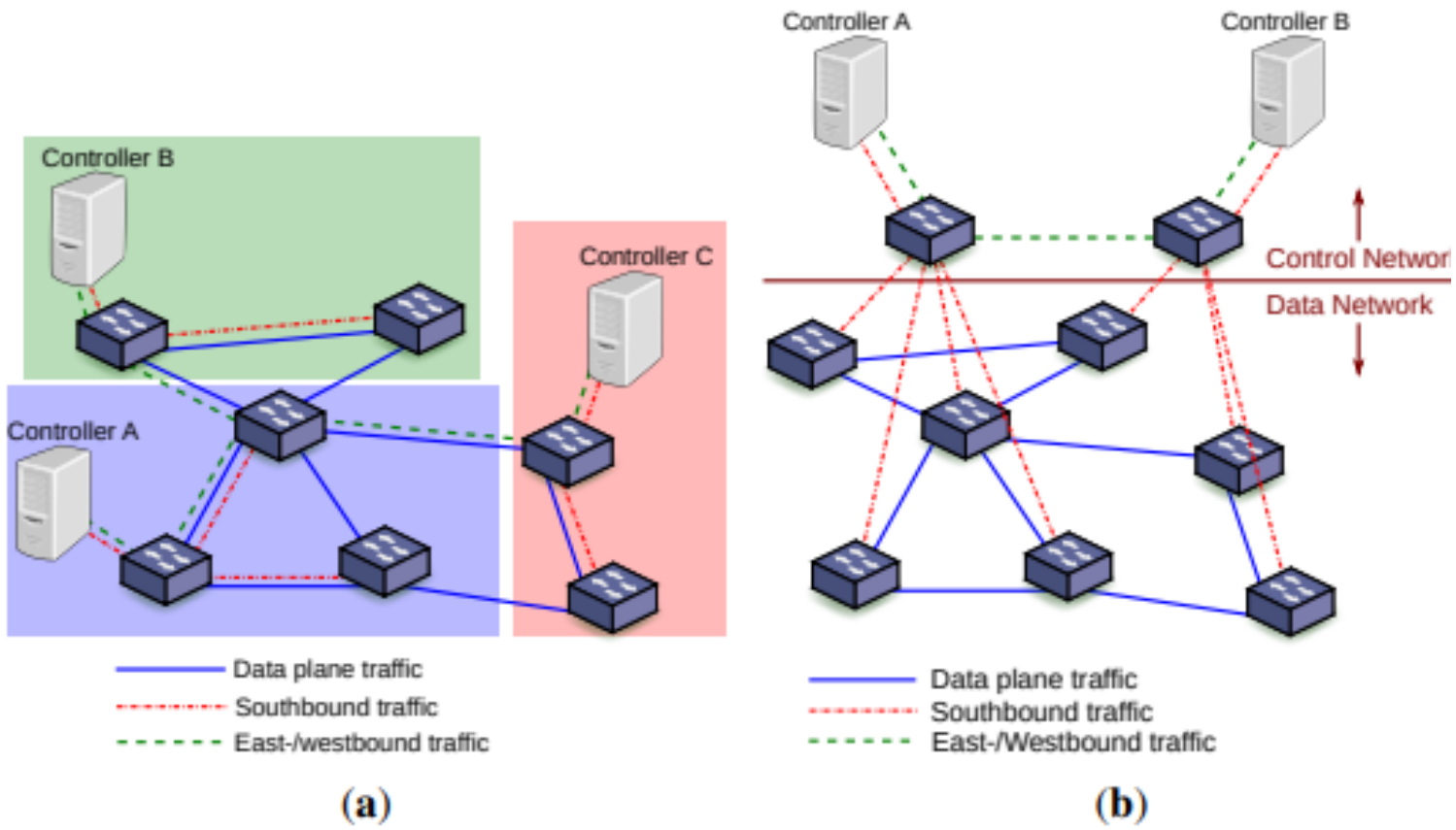
Centralized/Distributed Control

- **"Onix: A Distributed Control Platform for Large-scale Production Networks"**



Control plane signaling

Figure 9. (a) In-band signaling; (b) Out-of-band signaling.



Source: Software-Defined Networking Using OpenFlow: Protocols, Applications and Architectural Design Choices, Future Internet

OpenFlow Cont..

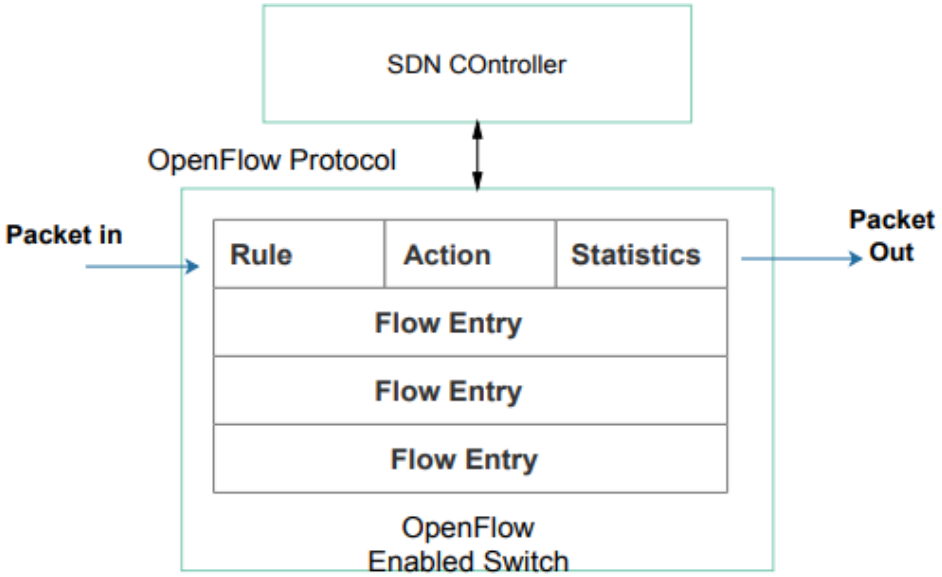


Figure 4: An abstract model of an OpenFlow switch.

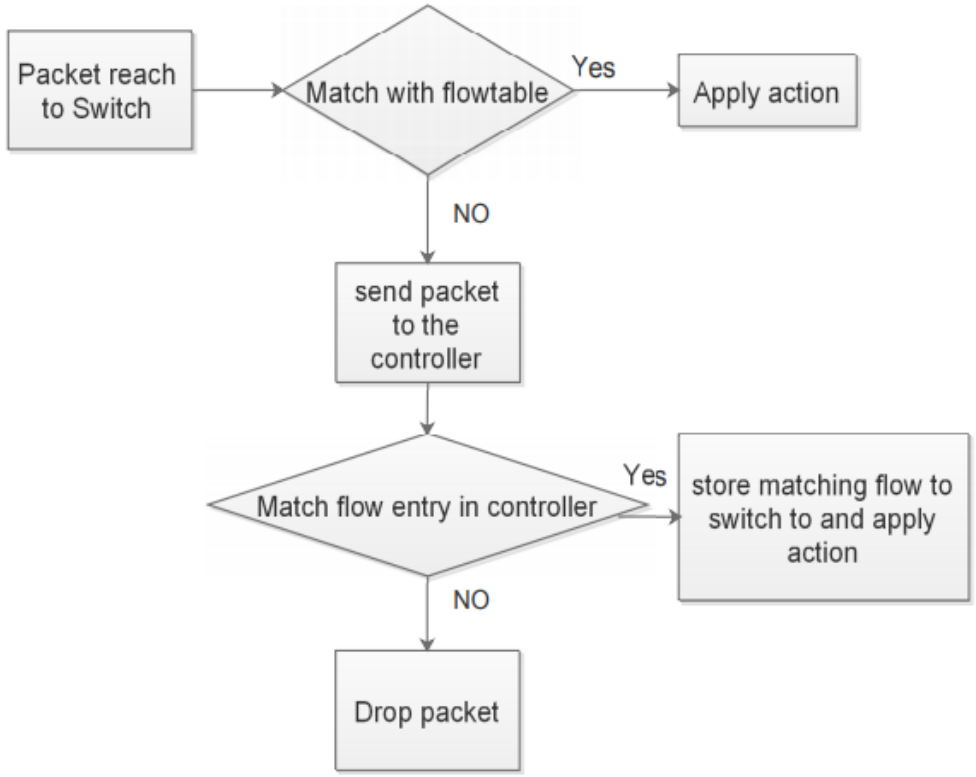


Figure 5: Flowchart of execution sequence of a packet in SDN.

Connection Est.

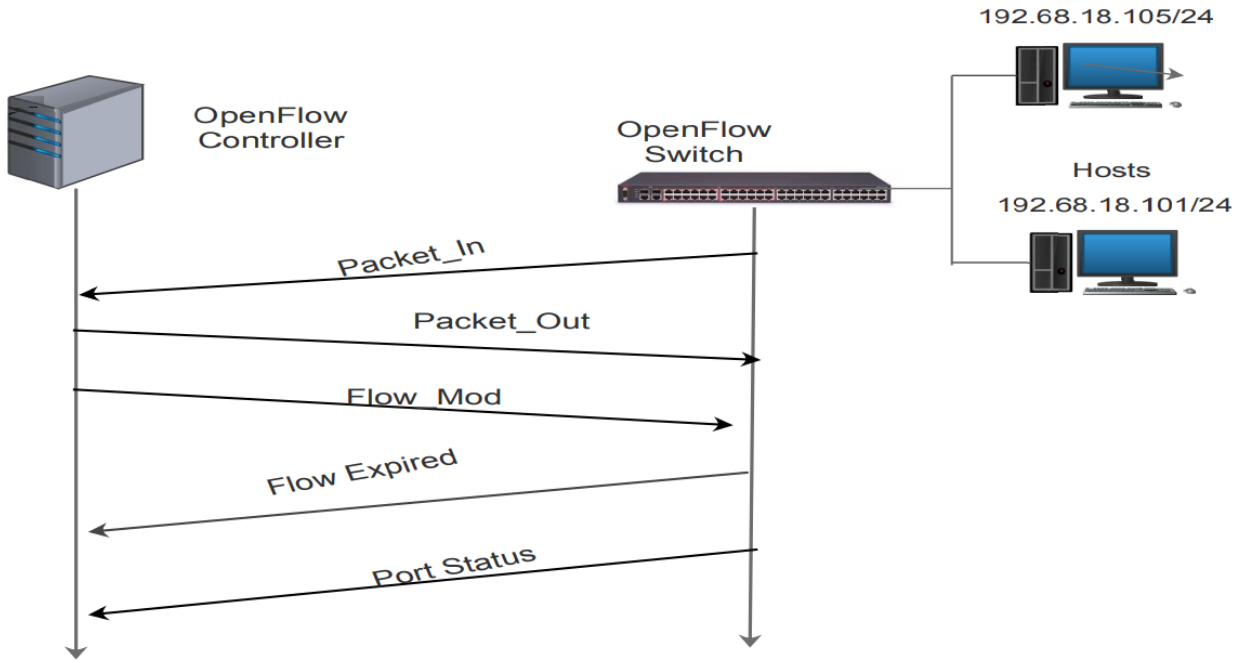
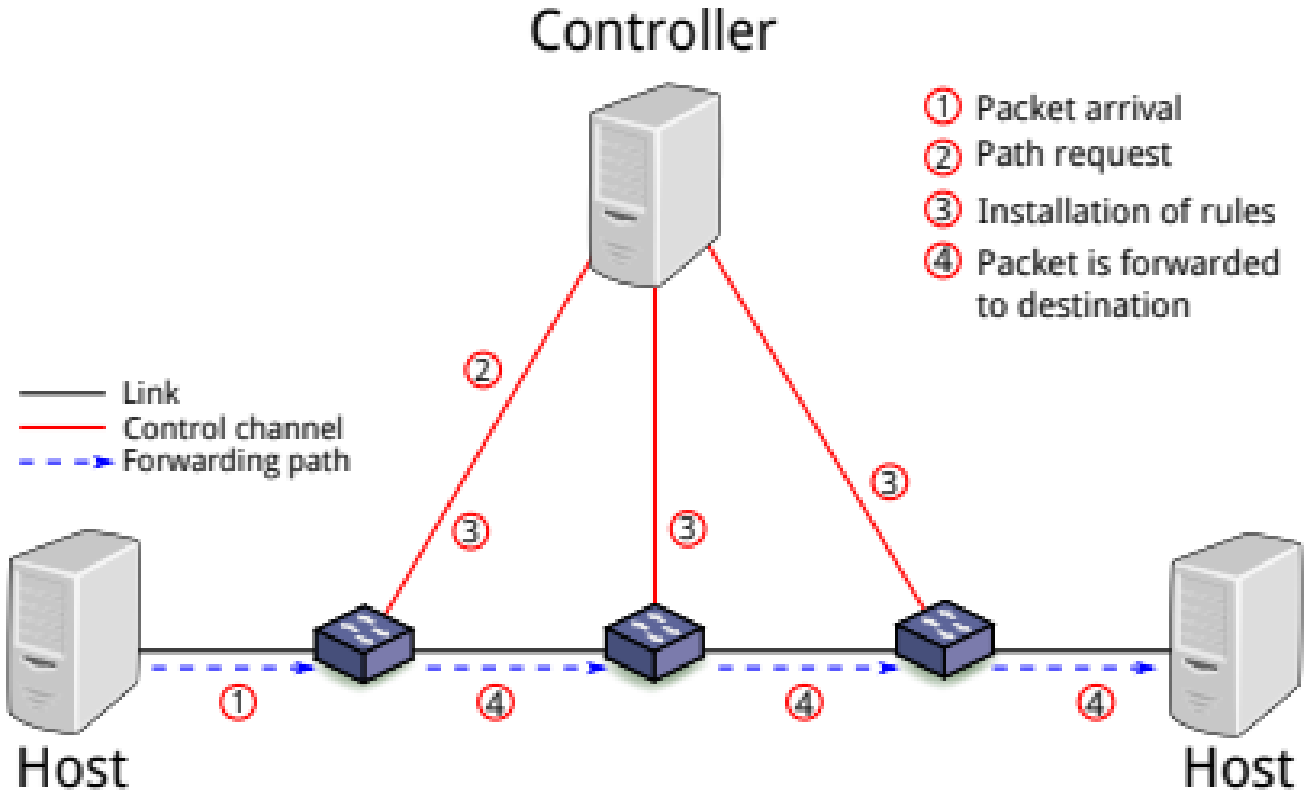


Figure 6: Connection establishment between hosts and the OpenFlow network.

Flow Management

Figure 10. Reactive flow management.



Source: Software-Defined Networking Using OpenFlow: Protocols, Applications and Architectural Design Choices, Future Internet

Very Good Survey Paper on SDN

- Software-Defined Networking: A Comprehensive Survey by Diego Kreutz and others [Added in the google classroom]